

AWS CRASH COURSE

1. What is AWS?

AWS (Amazon Web Services) is the world's largest cloud computing platform. It provides more than 200 cloud services that help individuals and companies build, deploy, and manage applications without owning physical servers.

AWS = Amazon Web Services.

AWS provides many services (compute, database, storage, DevOps, AI, etc.).

Companies use AWS to deploy applications.

AWS has data centers around the world.

Users access AWS resources through the internet.

2. Why was AWS created?

Before AWS:

- **Companies had to purchase servers.**
- **Build data centers.**
- **Hire maintenance teams.**
- **Upgrade hardware whenever traffic increased.**

This process was expensive and time-consuming.

AWS solved this problem by allowing users to rent computing resources over the internet.

3. What is cloud computing

Cloud computing is the delivery of computing resources such as servers, storage, databases, and networking over the internet instead of buying and maintaining your own physical infrastructure.

Cloud computing means you **don't own the physical server**.

You **rent computing resources** (server, storage, database, etc.) from AWS.

You access and manage everything **through the internet**.

AWS maintains the hardware; you focus on your application.

"I rent a virtual server from AWS."

4. What is Server

A server is a computer that provides services, data, or applications to other computers (clients) over a network or the internet. (cloud server)

A server is a computer.

It has CPU, RAM, storage, and other hardware.

It processes requests from users or applications.

It can store applications and data.

Login Process

Click the New signup option and they will ask you about your details and also Step 1 to 4 (includes : personal info, payment of 2rs and setting up autopay and fetching the aadhar through digilocker)

Login as root user(owner for doing the all services freely provided by aws)

EC2

EC2 is an AWS service that lets you create and run virtual servers (called EC2 instances) in the cloud.

EC2 = Elastic Compute Cloud.

EC2 is used to create a virtual server.

While creating it, you choose things like CPU, RAM, storage, operating system, and create a key pair for secure login.

The price depends on the resources you choose.

An EC2 instance is a virtual server running on AWS.

In the virtual computer we can increase the RAM , change in CPU, Increase in Storage so then AWS will ask for pay accordingly

When creating an server using ec2 many different services may interfere and joined with the ec2 services like

EC2 - service for creating server

EBS - service for storage

ROUTE 53 - service for DomainName

VPC - Network

Different services are there in AWS FOR different purposes

CREATING AN SERVER IN EC2 INSTANCE STEP BY STEP PROCESS

STEP 1 : CLICK THE SERVICES ICON IN AWS AND CLICK THE COMPUTE

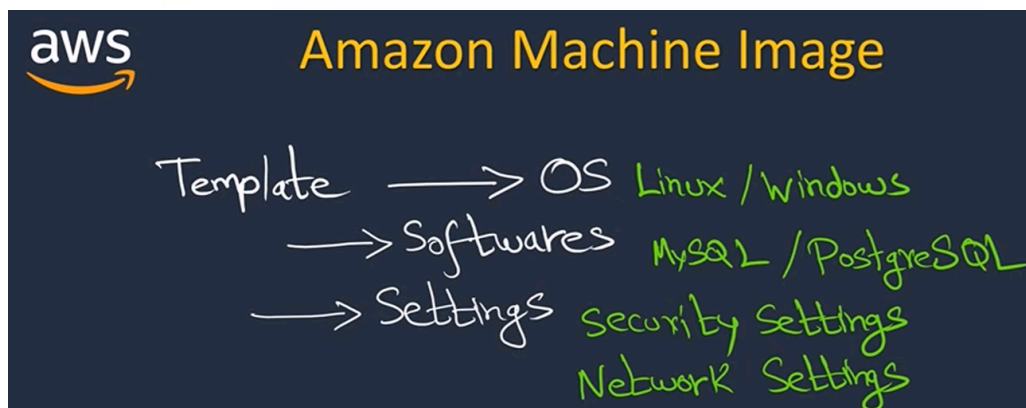
STEP 2 : THERE WILL BE IN OPTION CALLED EC2 CLICK THIS AND IT WILL REDIRECT TO THE CORRESPONDING PAGE

STEP 3 : Click the Instances and click launch instances

1. Set up a name for the server

2. Set up a AMAZON MACHINE IMAGE

The server should must contains an Any one OS(LINUX, REDHAT, WINDOWS, DEBIAN ETC...)



An Amazon Machine Image (AMI) is a pre-configured template that contains an operating system and required software used to launch an EC2 instance.

AMI stands for Amazon Machine Image.  AMI = Which operating system do you want?

It contains an operating system (Amazon Linux, Ubuntu, Windows, etc.).

When you launch an EC2 instance, AWS installs the selected AMI on that virtual server.

The operating system allows you to interact with the server.

The AMI already contains the operating system. When you launch an EC2 instance, AWS creates the virtual server using that AMI.

✓ If you choose Amazon Linux or Ubuntu:

- You connect using SSH.
- You use Linux commands (`ls`, `cd`, `pwd`, `mkdir`, etc.).

✓ If you choose Windows Server:

- You usually connect using Remote Desktop (RDP).
- You get a Windows desktop interface, not just a terminal.
- You can also use Command Prompt or PowerShell inside Windows.

3. Instance Type

Instance Type defines the hardware resources of your virtual server. The pricing depends on the instance type you choose. An Instance Type defines the hardware configuration of an EC2 instance, including CPU, memory, storage, and networking performance.

- Instance Type = CPU + RAM + Network + (sometimes GPU)
- Price = Based on the instance type.

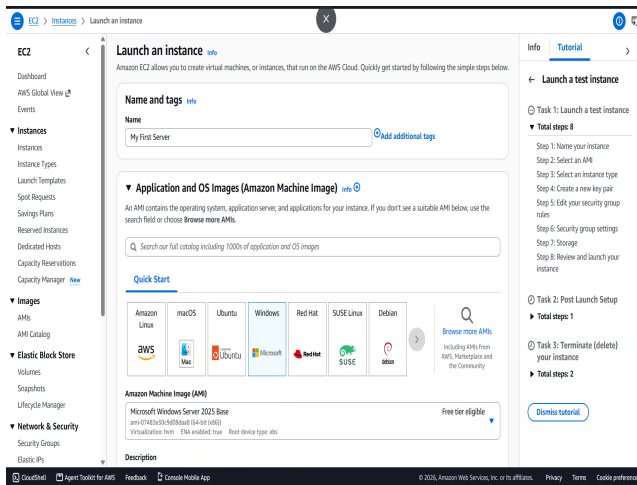
Instance Type determines the hardware configuration of your EC2 instance.

It includes:  Instance Type = How powerful should the server be?

- CPU (vCPU)
- RAM
- Network performance
- (Some instance types also include GPUs.)

You choose the instance type based on your application's requirements.

More powerful instance types generally cost more. The instances will be shown with prior to the selected “AMI IMAGE”



Instance Family	Main Resource	Example
Memory Optimized	More RAM	Databases, Caching
Compute Optimized	More CPU	Scientific computing, High-performance servers
Accelerated Computing	GPU	AI, Gaming, Video Rendering

4. Key Pair Value

A Key Pair is a security mechanism used to securely connect to an EC2 instance. It consists of a public key stored in AWS and a private key stored on the user's computer.

A Key Pair consists of two keys:

- Public Key → Stored by AWS on your EC2 instance.
- Private Key → Downloaded to your computer (.pem or .ppk).

1. Key Pair Name - This is just the name you give & It is only for identifying your key pair.

2. Key Pair Type - RSA vs Ed25519

Both are cryptographic algorithms used to create EC2 Key Pairs for secure SSH login.

How Key Pair Works:

- **Public Key** → Stored on the EC2 instance.
- **Private Key** → Downloaded to your computer.
- During login, AWS verifies that both keys match before granting access.

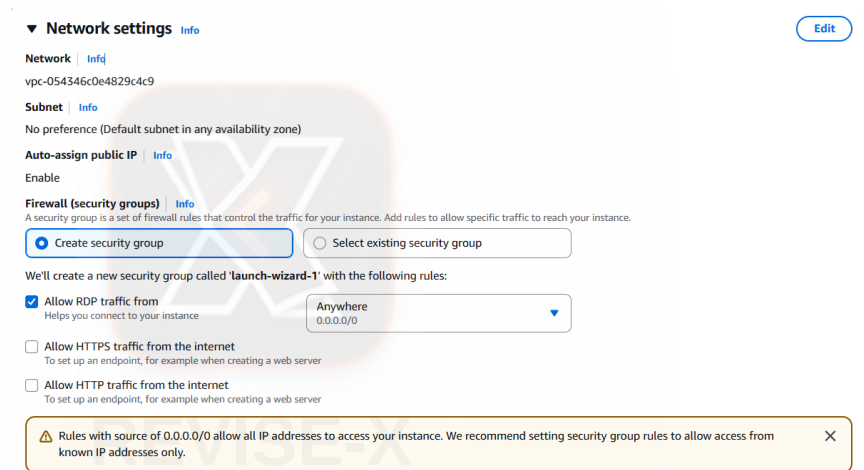
Difference

- **RSA** → Older, widely supported, larger key size.
- **Ed25519** → Newer, faster, smaller key size, and more secure.

3. Private Key File Format - .pem vs .ppk

- **.pem** → Used with OpenSSH (Linux, macOS, Windows Terminal).
- **.ppk** → Used with PuTTY (Windows).

5. NetWork Security



▼ Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-054346c0e4829c4c9

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow RDP traffic from Helps you connect to your instance Anywhere 0.0.0.0/0

☐ Allow HTTPS traffic from the internet To set up an endpoint, for example when creating a web server

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⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Network Settings configure how your EC2 instance connects to the network and the internet (VPC, Subnet, Public IP, etc.).

Security Groups act as a firewall. They decide who can access your EC2 instance and through which ports/protocols (SSH, HTTP, HTTPS, RDP, etc.). Security Groups decide who can access your EC2 instance (server), not your AWS account.

Network (VPC) — Creates a private network for your AWS resources.

Subnet — Divides the VPC into smaller networks and places the EC2 instance in one of them.

Auto-Assign Public IP — Assigns a public IP so the EC2 instance can be accessed over the internet.

Create Security Group — Creates a new firewall for your EC2 instance.

Select Existing Security Group — Uses an already created firewall.

Allow RDP Traffic (Port 3389) — Allows Remote Desktop access to a Windows server.

Allow HTTP Traffic (Port 80) — Allows users to access your website using HTTP.

Allow HTTPS Traffic (Port 443) — Allows secure website access using HTTPS.

0.0.0.0/0 (Anywhere) — Allows connections from any IP address on the internet.

5. Storage Configuration

Configure Storage is used to allocate the storage space (disk space) for the EC2 instance based on the application's requirements.

▼ **Configure storage** [Info](#) Advanced

1x GiB Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

🔄 Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

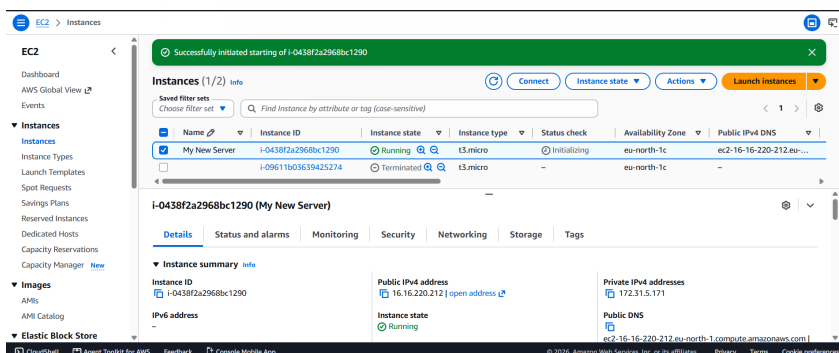
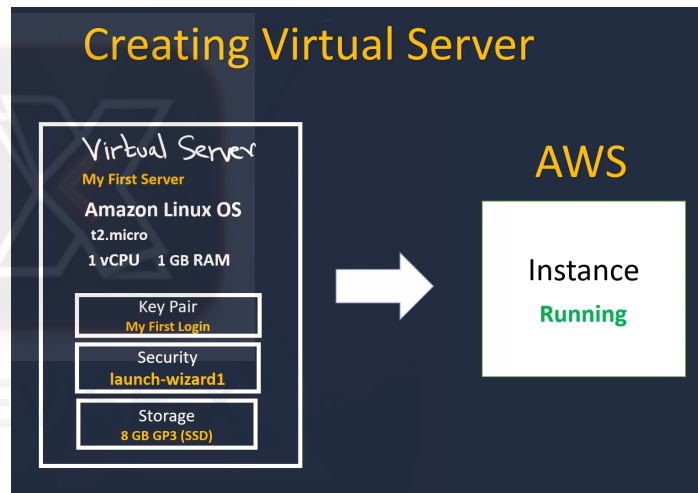
File systems

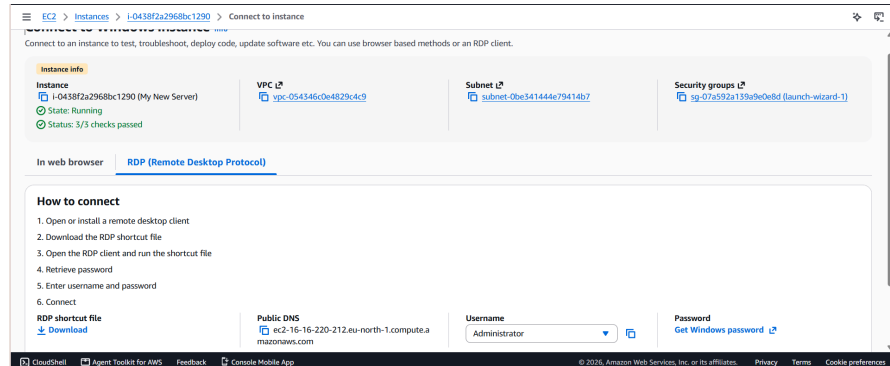
☐ S3 Files - new
S3 Files is not supported on Windows instances.

☐ EFS
EFS is not supported on Windows instances.

☐ FSx

☒ None





AWS EC2 Windows – Remote Desktop Connection

Step 1: Launch EC2

- Create a Windows EC2 instance.
- Allow RDP (Port 3389) in the Security Group.

Step 2: Get Password

- Go to Connect → RDP Client.
- Upload the .pem file.
- Click Decrypt Password.
- Copy the Administrator Password.

Step 3: Download RDP File

- Click Download Remote Desktop File (.rdp).

Step 4: Open the RDP File

- Double-click the .rdp file.
- If a warning appears, click Connect.

Alternative Method (If .rdp doesn't open)

1. Press Win + R.
2. Type mstsc and press Enter.
3. Click Show Options → Open.
4. Select the downloaded .rdp file.

5. Click Connect (the public ip of the instance change when ever the server is starting so use the crt public ip address after every restarting of the server(ec2))

Step 5: Login

- Username: **Administrator**
- Password: Paste the password copied from AWS.
- Click OK.

Step 6: Accept the Warning

- Click Yes to continue.

Step 7: Connected Successfully

- Your Windows EC2 desktop will open.



Simple Storage Service - It is an cloud storage used to store and retrieve the data

Its use cases are : Data Backup , web hosting , data archiving

S3 is used to store data.

You can upload files (objects).

You can download/retrieve files.

S3 is highly secure and durable.

S3 supports archiving for files that are rarely accessed.

1.BUCKET

An S3 bucket is a container that stores objects (files) in the AWS cloud.

Bucket is like a folder.

Files stored inside a bucket are called objects.

You upload objects into a bucket.

A bucket is used to organize and store data.

The object in the bucket can be a audio , video , image any data(various file format)

2.Storage Classes

S3 storage has many standards like

- i) Standard - huge cost due to frequent usage of the objects(files)
- ii) Intelligent Tiering - reduced cost (keeps the most and daily using file in expensive storage and not used file in cheap storage(this itself will divide the files according to the usage))
- iii) glaciers - cost efficient

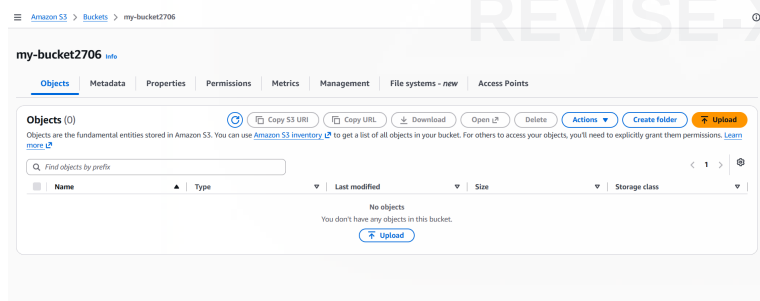
3.Security

Bucket policy - Rules for accessing the buckets

IAM user - Manage Permissions(*Who* is accessing AWS.)

Creating an bucket

- 1.Region → Where the bucket is created.
- 2.Bucket Name → Unique bucket name.
- 3.Object Ownership → Who owns uploaded files.
- 4..Block Public Access → Protects the bucket from public access.
- 5.Versioning → Tracks file changes.
- 6.Encryption → Secures stored data.
- 7.Object Lock → Prevents deletion or modification.



Uploading and downloading files in bucket

Buckets

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (1) [Info](#)



Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >



Name	AWS Region	Creation date
my-bucket2706	Europe (Stockholm) eu-north-1	August 7, 2026, 21:04:45 (UTC+05:30)

my-bucket2706 [Info](#)

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (1)



Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >



☐	Name	Type	Last modified	Size	Storage class
☐	Sample/	Folder	-	-	-

biodata.txt [Info](#)

Copy S3 URI

Download

Open

Object actions

Properties

Permissions

Versions

Object overview

Public Access to objects

The object(file) in the s3 bucket can be publicly accessible using an url by remove the block and we have to add access to everyone

Object Ownership

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☐ ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

Enabling ACLs turns off the bucket owner enforced setting for Object Ownership. Once the bucket owner enforced setting is turned off, access control lists (ACLs) and their associated permissions are restored. Access to objects that you do not own will be based on ACLs and not the bucket policy.

☐ I acknowledge that ACLs will be restored.

Edit Block public access (bucket settings)

Block public access (bucket settings)

Public access to objects in buckets can be blocked through access control lists (ACLs), bucket policies, access point policies, or all, in order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to tailor your specific storage use case. [Learn more](#)
☐ Block of public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly created buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to resources using ACLs.

☐ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will prevent all ACLs that grant public access to buckets and objects.

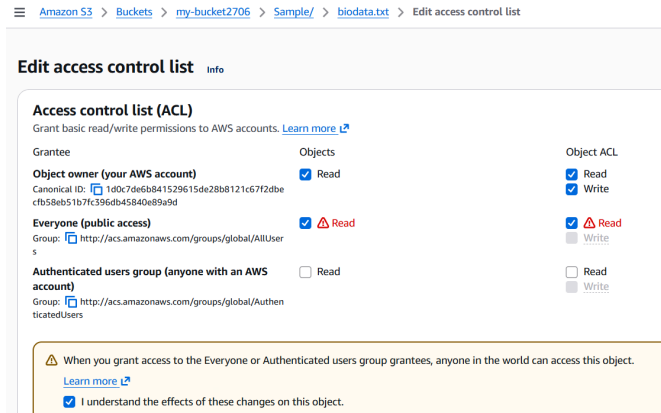
☐ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through any public bucket or access point policies

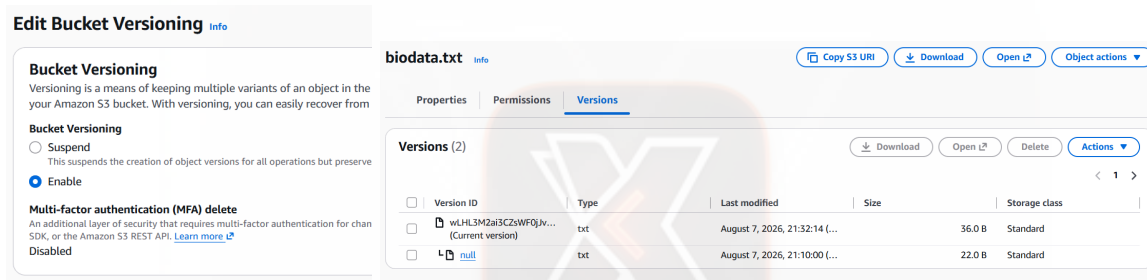
S3 will prevent public and cross-account access to buckets or access points with policies that grant public access to buckets and objects.

Cancel Save



Versions :

Bucket Versioning – Stores multiple versions of an object to track changes/modifications and recover deleted or previous versions of the file.



S3 PERMISSIONS :

It defines that what the user can do with an S3 bucket.

For accessing the buckets and objects then the IAM Users have to be created

IAM (Identity and Access Management) :

IAM is an AWS service used to create and manage users and their permissions.

It decides the Who can view a bucket. Who can upload files. Who can download files. Who can delete files.

The IAM Is an service for creating the user to access , modify the created bucket



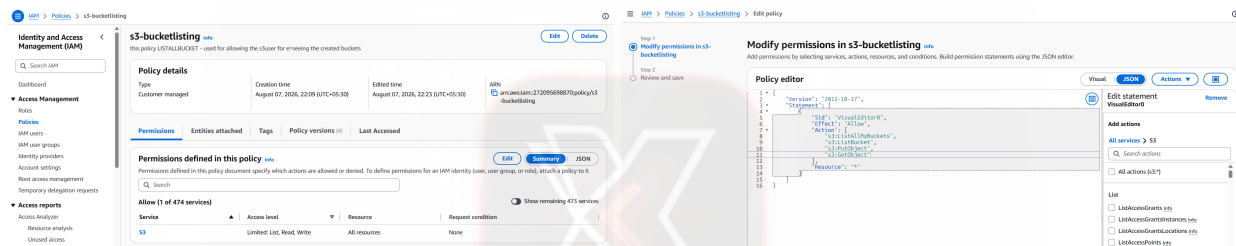
For giving the permission for the IAM user to view the created bucket then in the Root User account go to > IAM > POLICIES > CREATE POLICIES > THEN CLICK "LISTALLBUCKETS"

After this process go to IAM>IAM USERS>S3 Users>ADD POLICIES.AND SELECT THE CUSTOM POLICY

After this refresh the iam user page and then the general bucket will be viewed but not the objects inside it

For viewing the object and opening the objects and downloading in the s3 buckets needs to enable some policies like

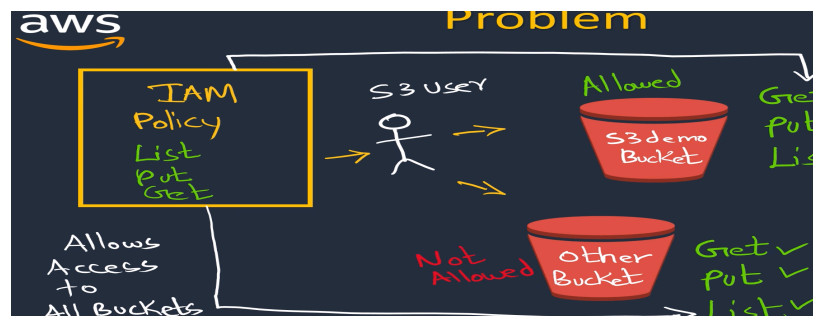
s3:ListAllMyBuckets, s3:ListBucket, s3:PutObject, s3:GetObject



BUCKET POLICIES

Attached to a specific S3 bucket. It says who can access this bucket and what they can do.

- Bucket policies are applied to one specific bucket.
- You can allow or deny access to that particular bucket.
- Bucket policies are written in JSON.
- You specify who can access the bucket and what they can do.



Amazon S3 > Buckets > permission-bucket2706

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::272095698870:user/s3-user"
      },
      "Action": [
        "s3:GetObject",
        "s3:PutObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::permission-bucket2706",
        "arn:aws:s3:::permission-bucket2706/*"
      ]
    }
  ]
}
```

[Edit](#) [Delete](#) [Copy](#)

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Amazon S3

Buckets

General purpose buckets **All AWS Regions** Directory buckets

General purpose buckets (2) [Info](#)

[Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
my-bucket2706	Europe (Stockholm) eu-north-1	August 7, 2026, 21:26:52 (UTC+05:30)
permission-bucket2706	Europe (Stockholm) eu-north-1	August 7, 2026, 22:41:02 (UTC+05:30)

[Account snapshot](#) [Info](#)
Updated daily
[View dashboard](#)
Storage Lens provides visibility into storage usage and activity trends.

[External access summary](#) [Info](#)
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

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